

Beyond Best-Single: Empirical Multi-Model Routing Across Accuracy, Cost, and Routing Overhead

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Abstract—Large language model routing attempts to preserve response quality while reducing inference cost by selecting a model for each query. This work evaluates a cost aware GPT 5 relative router under a leakage controlled protocol built from LLMRouterBench. Router design uses 3,450 development queries from four benchmarks across 13 models. A fresh confirmation cohort is frozen from five previously unused benchmark families, yielding 7,853 aligned queries. The final router embeds each prompt with BAAI bge m3, predicts GPT 5 relative correctness and cost signals for 12 alternatives, and offloads only when estimated accuracy risk remains within a development frozen tolerance and predicted dollar saving is positive. On 7,104 binary scored confirmation queries, GPT 5 achieves 39.626 percent accuracy and the router achieves 39.428 percent. The paired bootstrap interval ranges from negative 0.380 to negative 0.014 percentage points, providing evidence of a small accuracy loss, but its lower bound remains within the pre specified 0.569 point retention margin. Across all 7,853 confirmation queries, recorded inference cost falls by 1.034 percent, with a 95 percent interval from 0.642 to 1.498 percent. The offload rate falls from 10.4 percent in development to 1.885 percent in confirmation, exposing substantial transport shift. RouteLLM saves more cost but incurs a larger accuracy loss, while Avengers Pro trades a large accuracy loss for much larger savings. Local routing overhead is 46.7 milliseconds at the median and becomes several seconds on long SWE Bench prompts. The results support a narrow confirmed cost saving, not accuracy equivalence, and show that transport and routing overhead are central deployment constraints.

Index Terms—large language models, model routing, inference optimization, cost-aware inference, leakage control, paired evaluation

I. INTRODUCTION

LLM systems increasingly face a per-query model-selection problem. Candidate models differ in capability, specialization, and price, so a fixed strongest model can waste resources while a fixed cheaper model can lose quality. RouteLLM formalizes this quality-cost trade-off for a strong/weak pair [2]; GraphRouter, Avengers-Pro, and RouterDC extend routing to richer model sets and representations [3]–[5]. LLMRouterBench provides a common benchmark and reports substantial model complementarity, but it also shows that practical routers remain far from hindsight upper bounds [1].

A central empirical difficulty is that router development can easily contaminate evaluation. Thresholds, model pools,

baseline adaptations, and even metric interpretations may be revised after observing a nominal holdout. This is especially problematic when the desired effect is small. A router that saves only a few percent in cost can appear favorable or unfavorable because of a handful of discordant queries or because a threshold was implicitly tuned against held-out outcomes.

This study therefore treats evaluation protocol as a first-class part of the routing problem. An earlier 863-query outer partition had already been observed during a proof-of-concept phase, so it is explicitly excluded from all new V2 model, threshold, feature, and hyperparameter decisions. V2 is developed only on the original 3,450-query development partition. A separate 7,853-query confirmation cohort is constructed from previously unused LLMRouterBench datasets, query identities and router assignments are frozen before outcome access, and the primary accuracy and cost decision rules are fixed in advance. The fresh cohort is materially different and harder than the development corpus: it is dominated by HLE, SimpleQA, and SWE-Bench, and GPT-5 reaches only 39.626 percent binary accuracy. Absolute accuracy therefore should not be compared directly with the much higher values reported on the development-family benchmarks.

The contributions are fourfold. First, the work defines a GPT-5-relative router that jointly estimates correctness risk and dollar saving using only pre-inference prompt features. Second, it constructs an out-of-fold accuracy-cost frontier and selects a development operating point under a pre-frozen accuracy-retention rule rather than a post-hoc point estimate. Third, it performs one fresh common-protocol confirmation against GPT-5, RouteLLM, and Avengers-Pro. Fourth, it measures local router overhead separately from model-generation latency, allowing deployment cost and routing overhead to be discussed without claiming unmeasured end-to-end speedup.

II. RELATED WORK

RouteLLM learns a router from preference data and selects between a strong and weak model at a configurable threshold [2]. GraphRouter represents tasks, queries, and LLMs as a heterogeneous graph and predicts query-model edge utility

TABLE I
DEVELOPMENT AND FRESH CONFIRMATION COHORTS

Phase / Dataset	Queries	Role
AIME	48	Development
GPQA	158	Development
LiveCodeBench	844	Development
MMLU-Pro	2400	Development
Development total	3450	Fit/select
LiveMathBench	121	Confirmation
SWE-Bench Verified	500	Confirmation
HLE	2157	Confirmation
SimpleQA	4326	Confirmation
Arena-Hard	749	Confirmation
Confirmation total	7853	One-time test

[3]. Avengers-Pro clusters query embeddings and ranks models using a performance-efficiency objective [4]. RouterDC learns query and model representations through dual contrastive objectives [5]. These systems cover pairwise preference routing, graph prediction, cluster routing, and contrastive retrieval.

LLMRouterBench unifies multiple routing datasets, model pools, and performance-cost settings [1]. The development corpus used here includes AIME, GPQA [7], LiveCodeBench [8], and MMLU-Pro [9]. Fresh confirmation uses LiveMathBench [11], SWE-Bench [12], Humanity’s Last Exam (HLE) [13], SimpleQA [14], and Arena-Hard [15]. BGE-M3 supplies the 1,024-dimensional prompt representation [6]. Linear estimators are implemented with scikit-learn [10].

III. LEAKAGE-CONTROLLED EVALUATION PROTOCOL

A. Development and Confirmation Cohorts

The model pool contains 13 aligned systems, including GPT-5 and 12 alternatives. The V2 development set contains 3,450 queries from the previously constructed 4,313-query corpus. The already-observed 863-query V1 outer partition is retained only as legacy evidence and is not reused for V2 selection. Five-fold development cross-fitting is stratified by dataset with shuffle and seed 42, so each development query is scored by estimators that did not train on that query.

The fresh confirmation cohort is drawn from five LLM-RouterBench datasets that were not used for V2 fitting or threshold selection. Common-query intersection filtering is performed before outcome access so every retained query has all 13 model observations. HLE contains 2,158 records per model but non-identical query sets, producing a 2,157-query common intersection; Arena-Hard similarly yields 749 common queries from 750-record model files. The final cohort contains 7,853 queries, as summarized in Table I.

B. Outcome Sealing and Metric Semantics

The confirmation query hashes, prompt embeddings, V2 assignments, RouteLLM assignments, Avengers-Pro assignments, and final baseline roster were saved and hashed before fresh outcome values were opened. No threshold, estimator, assignment, baseline, or margin was changed after confirmation access.

Source-code inspection before outcome access established that LiveMathBench, SWE-Bench, HLE, and SimpleQA yield

binary correctness for the retained protocol. These 7,104 queries form the primary accuracy analysis. Arena-Hard instead preserves the benchmark’s native comparative score $\{0, 0.5, 1\}$, so its 749 queries are analyzed separately rather than post-hoc binarized. Recorded-cost confirmation uses all 7,853 queries.

For binary accuracy, let

$$A = \frac{1}{N} \sum_{i=1}^N z_i, \quad (1)$$

where $z_i \in \{0, 1\}$. The primary paired accuracy statistic is

$$\Delta_A = A_{\text{router}} - A_{\text{GPT5}}. \quad (2)$$

The accuracy-retention margin is derived before confirmation from development data only. We take the smaller of GPT-5’s dataset-stratified bootstrap standard error on development (0.568656 percentage points) and the smallest adjacent accuracy gap among the top three fixed models (1.101449 points), yielding $\delta = 0.005686564292$ (0.568656 percentage points). The router passes the pre-frozen retention criterion when the lower endpoint of the two-sided 95% paired, dataset-stratified bootstrap interval satisfies

$$L_{95}(\Delta_A) > -\delta. \quad (3)$$

The margin is study-specific and should not be interpreted as a universal tolerance for LLM quality loss.

Total recorded cost is $C = \sum_i c_i$, and cost reduction relative to GPT-5 is

$$R_C = 1 - \frac{C_{\text{router}}}{C_{\text{GPT5}}}. \quad (4)$$

A confirmatory positive-saving claim requires $L_{95}(R_C) > 0$. Both accuracy and cost intervals use 20,000 paired bootstrap replicates, stratified by confirmation dataset, with seed 42. Exact McNemar testing is retained as a diagnostic for binary discordant pairs.

IV. PROPOSED GPT-5-RELATIVE ROUTER

A. Representation and Relative Heads

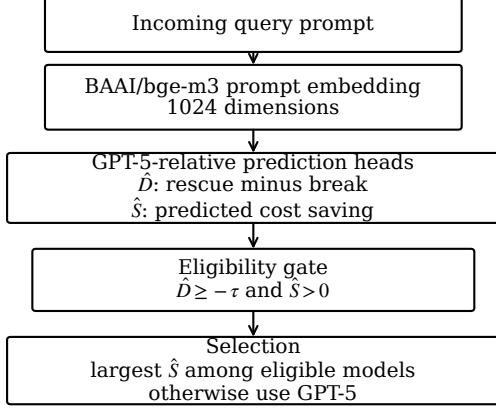
Each prompt is encoded with BAAI/bge-m3 into a 1,024-dimensional float32 vector with application-level normalization disabled. Dataset identity, model outputs, realized correctness, and realized candidate cost are not router inputs. For each alternative model m , an L2-regularized multinomial logistic head predicts three GPT-5-relative correctness outcomes: rescue, break, or tie. Define

$$\hat{D}(i, m) = \hat{P}(\text{rescue}) - \hat{P}(\text{break}). \quad (5)$$

A Ridge head with $\alpha = 1$ predicts the GPT-5-relative dollar saving

$$\hat{S}(i, m) = c_{\text{GPT5}}(\widehat{i}) - c_m(i). \quad (6)$$

The same estimator family is used in every cross-fit fold and in the final refit.



Only pre-inference prompt features enter the router.

Fig. 1. Frozen V2 routing pipeline. The risk budget τ is selected using development data only. No fresh outcome enters routing.

B. Accuracy-Cost Frontier and Operating Point

The V2 family permits alternative m when

$$\hat{D}(i, m) \geq -\tau \quad \text{and} \quad \hat{S}(i, m) > 0. \quad (7)$$

If no alternative qualifies, GPT-5 is selected. Otherwise the router chooses the eligible model with largest $\hat{S}$, breaking ties by larger $\hat{D}$ and then canonical model name.

Five-fold out-of-fold predictions over all 3,450 development queries generate 10,067 realized operating points and a 349-point Pareto frontier. At $\tau = 0$, the router reduces development cost by 2.31% with a -0.029-point accuracy difference. Seven Pareto points satisfy the frozen development accuracy rule. The lowest-cost passing point uses $\tau = 0.00537$ (full precision is retained in the frozen experimental artifact). It produces 359 development offloads, an accuracy difference of -0.203 points with a 95% interval of [-0.522, 0.087] points, and a 4.410% observed cost reduction. The paired cost interval is [3.155%, 5.741%]. These development intervals are selection-affected and are not treated as confirmation evidence. After selection, the same estimator family is refit on all 3,450 development queries and the resulting weights, τ , and routing rule are frozen before fresh outcomes are accessed.

V. BASELINES

GPT-5 is the fixed accuracy anchor. RouteLLM uses the frozen matrix-factorization checkpoint for GPT-5 versus Gemini 2.5 Flash with threshold 0.5. Direct checkpoint reconstruction reproduces all 863 legacy assignments before fresh application. Avengers-Pro is reconstructed with BGE-M3 embeddings, 32 clusters, top- $k = 1$, $\beta = 9$, and performance/cost weights 0.7/0.3; its reconstruction also exactly reproduces all 863 historical assignments before fresh routing. Fresh baseline assignments depend only on the already-frozen query representations and trained state.

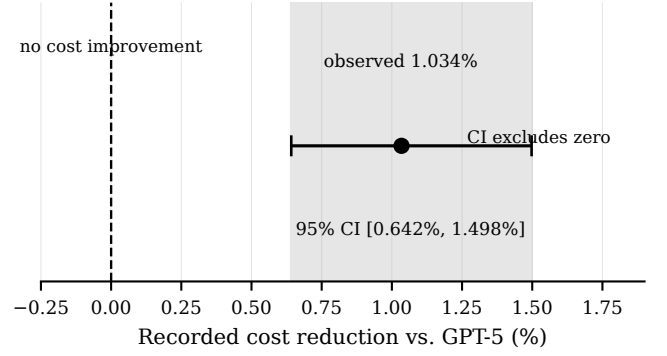


Fig. 2. Observed recorded cost reduction and its 95% paired percentile-bootstrap interval on the fresh confirmation cohort. The interval lies entirely above zero.

The common fresh baseline roster is therefore GPT-5, the proposed router, RouteLLM, and Avengers-Pro. GraphRouter remains a historical comparison because its frozen implementation uses task-description representations and per-dataset cost normalization fit for the four development tasks; applying it to five unseen task identities would require a new transport rule after development. RouterDC is not executed because the required local 7B artifact/runtime is unavailable. Max Expert remains a fixed historical comparator rather than a fresh query router. These exclusions were recorded before fresh outcomes were opened.

VI. FRESH CONFIRMATION RESULTS

A. Primary Accuracy and Cost Confirmation

Table II gives the common-protocol results. On the 7,104 binary-scored queries, GPT-5 answers 2,815 correctly and V2 answers 2,801 correctly. Accuracy therefore changes from 39.626% to 39.428%, or -0.197 percentage points. The paired 95% interval is [-0.380, -0.014] points. Although the interval is below zero, its lower endpoint remains above the pre-frozen -0.569-point boundary, so the accuracy-retention criterion passes. This is not an equivalence claim: 14 GPT-5 errors are rescued, 28 GPT-5 successes are broken, and exact McNemar $p = 0.0436$, indicating evidence of a small paired accuracy disadvantage.

Across all 7,853 confirmation queries, GPT-5 costs \$250.959431 and V2 costs \$248.363575. The saving is \$2.595857, or 1.034%, with a paired 95% cost-reduction interval of [0.642%, 1.498%]. Because the lower bound is positive, the cost criterion passes. Under the pre-frozen joint rule, V2 therefore satisfies the study's combined accuracy-retention and positive-cost-saving confirmation criteria.

B. Macro, Arena-Hard, and Per-Dataset Behavior

Across the four binary datasets, GPT-5 macro accuracy is 42.342% and V2 macro accuracy is 42.226%, a -0.116-point difference with a 95% interval of [-0.209, -0.029] points. This secondary statistic is consistent with a small distributed accuracy cost rather than exact preservation.

TABLE II
FRESH COMMON-PROTOCOL RESULTS

System	Bin. Acc. %	Δ Acc. pp	Arena	Cost \$	Cost Red. %	Loss/Save
GPT-5	39.626	0.000	0.6976	250.9594	0.000	–
Proposed	39.428	-0.197	0.6983	248.3636	1.034	0.191
RouteLLM	39.091	-0.535	0.6896	247.0085	1.574	0.340
Avengers-Pro	29.575	-10.051	0.6909	143.0427	43.002	0.234

Binary accuracy uses 7,104 queries from LiveMathBench, SWE-Bench, HLE, and SimpleQA. Arena is the mean native Arena-Hard score on 749 queries. Cost covers all 7,853 confirmation queries. Loss/Save is the absolute binary-accuracy loss in percentage points divided by each percentage point of recorded cost reduction; it is descriptive, not confirmatory.

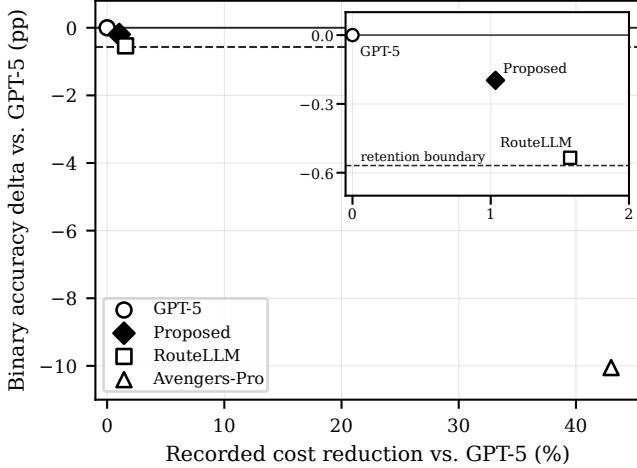


Fig. 3. Fresh accuracy-cost trade-off. The inset magnifies the 0–2% cost-reduction region containing GPT-5, V2, and RouteLLM. The dashed line is V2’s pre-frozen accuracy-retention boundary.

Arena-Hard is analyzed in its native $\{0, 0.5, 1\}$ space. GPT-5 has mean score 0.697597 and V2 has 0.698264. The paired mean difference is 0.000668 with a 95% interval of $[-0.003338, 0.005340]$, so no directional Arena-Hard improvement is established.

V2 makes 148 offloads in the full cohort, only 1.885% of queries. It retains GPT-5 on 7,705 queries (98.115%). Offloads are concentrated in SimpleQA (96), HLE (37), Arena-Hard (11), and LiveMathBench (4); SWE-Bench receives no offloads. In that same order, the recorded cost reductions are 1.957%, 0.919%, 0.662%, and 0.669%; SWE-Bench remains at 0%, as shown in Fig. 4.

C. Transport Shift Across Benchmark Families

The operating point changes sharply when transported away from the development benchmark families. On development, V2 offloads 359 of 3,450 queries (10.41%) and reduces recorded cost by 4.410%, with a selection-affected paired interval of $[3.155\%, 5.741\%]$. On fresh confirmation, it offloads only 148 of 7,853 queries (1.885%) and reduces cost by 1.034%, with interval $[0.642\%, 1.498\%]$. Thus the offload rate falls by more than a factor of five and the observed cost effect by roughly a factor of four; the entire confirmation interval also lies below the development interval. Because the cohorts contain different benchmark families, this is not a

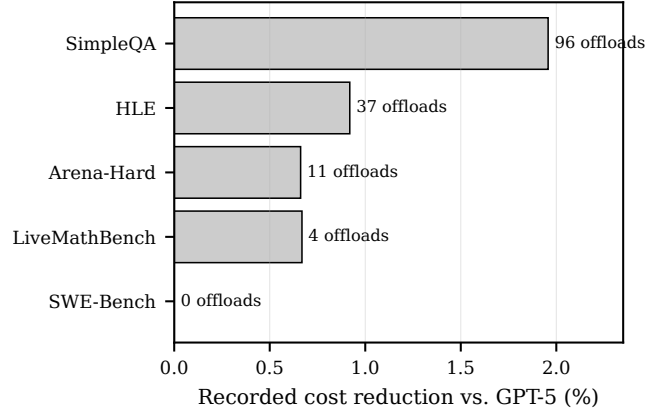


Fig. 4. Proposed-router recorded cost reduction by fresh confirmation dataset, ordered to match the discussion in text. Frozen offload counts are annotated.

formal paired cross-cohort test. It is, however, direct evidence that the development frontier did not transport quantitatively. This shift is precisely why the previously observed 863-query partition was not reused as confirmation evidence.

D. Secondary Baseline Trade-offs

RouteLLM selects GPT-5 for 7,605 queries and Gemini 2.5 Flash for 248. It saves 1.574% recorded cost but loses 0.535 binary-accuracy points relative to GPT-5, a larger point loss than V2. Its point estimate lies only 0.034 points inside V2’s -0.569-point boundary, but no baseline retention test was pre-specified; the point estimate alone therefore does not establish that RouteLLM satisfies V2’s interval-based criterion. Avengers-Pro offloads 4,106 queries and reduces cost by 43.002%, but its binary accuracy is 29.575%, 10.051 points below GPT-5. A descriptive efficiency ratio further separates the operating points: absolute accuracy loss per percentage point of cost reduction is 0.191 for V2, 0.340 for RouteLLM, and 0.234 for Avengers-Pro. V2 therefore has the smallest observed accuracy sacrifice per unit of recorded cost saving among the routed systems in Table II. Only the V2 versus GPT-5 accuracy and cost criteria were pre-frozen as confirmatory hypotheses; these baseline comparisons and ratios are secondary.

VII. ROUTER OVERHEAD

Local routing overhead is measured after confirmation without changing the frozen router. A deterministic 256-query sample is allocated proportionally across the five confirmation datasets. Each prompt is embedded batch-1 on Apple MPS with the same local BGE-M3 model, followed by the 12 correctness heads, 12 cost heads, and selection rule. Five dataset-specific warm-up queries are excluded. Batch-1 inference reproduces all 256 frozen assignments, with maximum absolute embedding difference 8.23×10^{-7} from the original frozen embeddings.

Table III shows that scoring and selection are inexpensive: the median is 1.35 ms and p95 is 1.76 ms. Prompt embedding

TABLE III
LOCAL BATCH-1 ROUTER OVERHEAD ON 256 QUERIES

Component	Median	P95 milliseconds	Mean
Prompt embedding	45.291	4175.691	343.836
Scoring + selection	1.348	1.755	1.435
Total router	46.734	4178.167	345.270

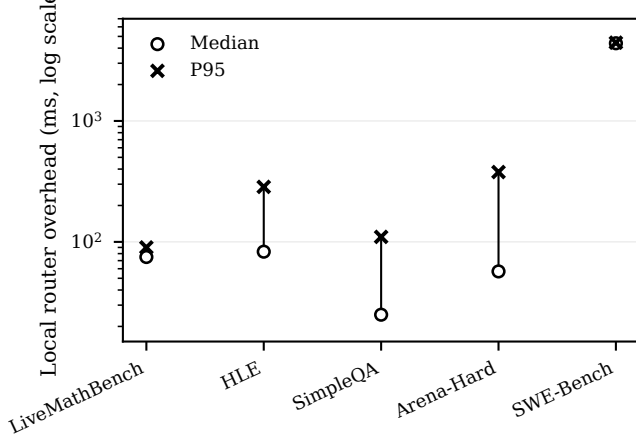


Fig. 5. Median and p95 local router overhead by dataset. The log scale exposes the long-prompt SWE-Bench tail.

dominates, producing a 46.73 ms median total router overhead. The overall p95 is 4.18 s because SWE-Bench prompts are extremely long: its sampled median is 19,741 raw tokens, 12 of 16 sampled prompts exceed BGE-M3’s 8,192-token maximum, and its median router overhead is 4.39 s. Because only 16 SWE-Bench prompts enter the proportional latency sample, extreme-tail quantiles are not stable; p99 is therefore omitted. More importantly, the frozen router makes zero SWE-Bench offloads, so on that workload the current policy pays several seconds of routing latency to select GPT-5 anyway.

The benchmark measures L_{router} only. A routed query improves end-to-end latency only when

$$L_{\text{GPT5}} - L_{\text{selected}} > L_{\text{router}}, \quad (8)$$

so a median case requires more than 46.7 ms of selected-model generation-time savings before routing breaks even on latency. Likewise, the observed model-cost saving is $2.595857/7853 = \$0.0003306$ per query; if local routing compute were monetized, its average per-query cost would need to remain below that amount to preserve the recorded model-cost advantage. Live provider timing and local-compute accounting are not available here, so these are break-even conditions rather than deployment claims.

VIII. THREATS TO VALIDITY AND LIMITATIONS

First, the confirmation cohort is fresh with respect to V2 design but is not an independent external benchmark collection: all datasets come from the same LLMRouterBench release and share its model outputs and cost recording conventions. The

outcome sealing is procedural and cryptographic rather than physical blinding.

Second, the primary binary cohort is highly imbalanced, with SimpleQA contributing 4,326 of 7,104 queries. Sample-weighted accuracy is therefore the primary statistic and macro-dataset accuracy is reported separately. Arena-Hard uses a different native metric and is not folded into binary accuracy.

Third, the 0.568656-point accuracy-retention margin is development-derived and study-specific. The confirmation interval excludes zero in the unfavorable direction, and McNemar testing also detects a small accuracy disadvantage. The correct interpretation is therefore that V2 met the pre-frozen retention tolerance, not that it is accuracy-equivalent to GPT-5.

Fourth, threshold selection uses a development Pareto frontier whose final cheapest-passing selection rule was formulated after inspecting development results. Cross-fitting removes direct in-sample scoring for each query, but development uncertainty remains selection-affected. The fresh cohort, not the development frontier, is the basis for the final claim.

Fifth, BGE-M3 truncation affects long prompts. Across the 7,853 fresh prompts, 347 exceed the model’s 8,192-token maximum, including 346 SWE-Bench prompts and one Arena-Hard prompt. No manual truncation rule is introduced; the frozen encoder’s native maximum-sequence behavior is used. The fresh router makes zero SWE-Bench offloads, but truncation still influences its routing scores and is a deployment limitation.

Sixth, the fresh common-protocol comparison contains only two learned routing baselines in addition to V2. GraphRouter could not be transported to unseen task identities without a new task-specific rule, and RouterDC could not be reproduced in the available environment, so baseline breadth remains limited.

Finally, costs are provider-reported values recorded by LLMRouterBench, not audited invoices. Model prices can change. The latency experiment measures local router overhead only and does not include network, provider queueing, time-to-first-token, or full generation time.

IX. CONCLUSION

This study combines a conservative GPT-5-relative router with a leakage-controlled evaluation protocol. V2 is selected entirely on development data, frozen, and then evaluated once on 7,853 previously unused LLMRouterBench queries. On 7,104 binary-scored queries, the router loses 0.197 accuracy points relative to GPT-5, but the 95% paired interval remains within the pre-frozen 0.569-point retention tolerance. Across the full cohort, recorded cost falls by 1.034%, and the 95% cost-reduction interval remains strictly positive. The joint pre-frozen confirmation rule therefore passes.

The result is intentionally narrow. It does not show accuracy equivalence, and it does not show end-to-end latency superiority. Instead, the fresh cohort detects a small accuracy loss alongside a small but statistically supported cost reduction. V2 retains the smallest observed accuracy loss per unit of cost saving among the routed systems, while RouteLLM buys

slightly more cost reduction with a larger loss and Avengers-Pro accepts a much larger quality drop for much larger savings. The fivefold reduction in offload rate from development to confirmation also shows that the router transports poorly across benchmark families, turning distribution shift into a central result rather than a footnote. Future work should improve transport across prompt distributions, avoid expensive routing on long-context workloads that rarely offload, evaluate dynamic pricing, and measure live selected-model latency under a common serving protocol.

REFERENCES

- [1] H. Li et al., “LLMRouterBench: A Massive Benchmark and Unified Framework for LLM Routing,” in *Findings of the Association for Computational Linguistics: ACL 2026*, San Diego, CA, USA, 2026, pp. 37733–37754, doi: 10.18653/v1/2026.findings-acl.1881.
- [2] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, T. Wu, J. E. Gonzalez, M. W. Kadous, and I. Stoica, “RouteLLM: Learning to Route LLMs from Preference Data,” in *Proc. 13th Int. Conf. Learn. Representations (ICLR)*, 2025.
- [3] T. Feng, Y. Shen, and J. You, “GraphRouter: A Graph-Based Router for LLM Selections,” in *Proc. 13th Int. Conf. Learn. Representations (ICLR)*, 2025.
- [4] Y. Zhang, H. Li, J. Chen, H. Zhang, P. Ye, L. Bai, and S. Hu, “Beyond GPT-5: Making LLMs Cheaper and Better via Performance-Efficiency Optimized Routing,” in *Proc. 2025 7th Int. Conf. Distributed Artificial Intelligence (DAI)*, London, U.K., 2025, pp. 122–129, doi: 10.1145/3772429.3772445.
- [5] S. Chen, W. Jiang, B. Lin, J. T. Kwok, and Y. Zhang, “RouterDC: Query-Based Router by Dual Contrastive Learning for Assembling Large Language Models,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 66305–66328, doi: 10.52202/079017-2120.
- [6] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation,” in *Findings of ACL 2024*, Bangkok, Thailand, 2024, pp. 2318–2335, doi: 10.18653/v1/2024.findings-acl.137.
- [7] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman, “GPQA: A Graduate-Level Google-Proof Q&A Benchmark,” in *Proc. 1st Conf. Language Modeling (COLM)*, 2023.
- [8] N. Jain, K. Han, A. Gu, W.-D. Li, F. Yan, T. Zhang, S. Wang, A. Solar-Lezama, K. Sen, and I. Stoica, “LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code,” in *Proc. 13th Int. Conf. Learn. Representations (ICLR)*, 2025.
- [9] Y. Wang et al., “MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 95266–95290, doi: 10.52202/079017-3018.
- [10] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *J. Mach. Learn. Res.*, vol. 12, no. 85, pp. 2825–2830, 2011.
- [11] J. Liu, H. Liu, L. Xiao, Z. Wang, K. Liu, S. Gao, W. Zhang, S. Zhang, and K. Chen, “Are Your LLMs Capable of Stable Reasoning?” in *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025, pp. 17594–17632, doi: 10.18653/v1/2025.findings-acl.905.
- [12] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. R. Narasimhan, “SWE-Bench: Can Language Models Resolve Real-World GitHub Issues?” in *Proc. 12th Int. Conf. Learn. Representations (ICLR)*, 2024.
- [13] Center for AI Safety, Scale AI, and HLE Contributors Consortium, “A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities,” *Nature*, vol. 649, pp. 1139–1146, 2026, doi: 10.1038/s41586-025-09962-4.
- [14] J. Wei, K. Nguyen, H. W. Chung, Y. J. Jiao, S. Papay, A. Glaese, J. Schulman, and W. Fedus, “Measuring Short-Form Factuality in Large Language Models,” arXiv:2411.04368, 2024.
- [15] T. Li, W.-L. Chiang, E. Frick, L. Dunlap, T. Wu, B. Zhu, J. E. Gonzalez, and I. Stoica, “From Crowdsourced Data to High-Quality Benchmarks: Arena-Hard and BenchBuilder Pipeline,” arXiv:2406.11939, 2024.